

Leveraging research into sex differences and steroid hormones to improve brain health

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Abstract

Sex differences, driven in part by steroid hormones, shape the structure and function of the brain throughout the lifespan and manifest across brain health and disease. The influence of steroid hormones on neuroplasticity, particularly in the adult hippocampus, differs between the sexes, which has important implications for disorders and diseases that compromise hippocampus integrity, such as depression and Alzheimer disease. This Review outlines the intricate relationship between steroid hormones and hippocampal neuroplasticity across the adult lifespan and explores how the unique physiology of male and female individuals can affect health and disease. Despite calls to include sex and gender in research, only 5% of neuroscience studies published in 2019 directly investigated the influence of sex. Drawing on insights from depression, Alzheimer disease and relevant hippocampal plasticity, this Review underscores the importance of considering sex and steroid hormones to achieve a comprehensive understanding of disease susceptibility and mechanisms. Such consideration will enable the discovery of personalized treatments, ultimately leading to improved health outcomes for all.

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Key points

- Sex should be consistently and meaningfully incorporated into research design and analysis to uncover possible sex influences on health and disease outcomes.
- Sex differences can be hidden as latent (or induced) differences or manifest in underlying mechanisms rather than in the initial trait itself.
- Sex-specific factors and experiences contribute to health and disease trajectories; within-sex comparisons are important beyond comparing to another sex in characterizing health outcomes.
- Steroid hormone fluctuations across the lifespan influence brain health and contribute to depression and Alzheimer disease risk; considering sex-specific effects in disease pathologies and treatments is crucial.
- Endogenous and exogenous steroid hormones exert sex-specific effects on the brain, including the ability to modulate adult hippocampal neurogenesis, depending on the dose, formulation, timing, genotype and experience.
- Adopting an intersectionality framework is necessary to understand the complex interplay of sex-specific factors and experiences that contribute to disease manifestation, progression and treatment response.

Introduction

Sex and gender influence the prevalence, manifestation and progression of, as well as the treatment response to, various diseases¹. Sex and gender are conceptually distinct and can interact in complex ways to influence health (Fig. 1). Sex refers to aspects of biological and physiological attributes, including chromosomes, gene expression, gonads and hormone levels in male, female and intersex individuals. Gender is a psychosocial construct that encapsulates gender identities, expressions, roles and expectations of girls, boys, women, men and gender-diverse people (such as transgender, gender non-conforming and two-spirit individuals). Gendered experiences can be driven by social, cultural and institutional influences that interact in varied ways and can have consequences on health that need to be thoughtfully examined^{2,3}. Neither sex nor gender is binary, despite often being treated as such in the literature. Sex influences are typically mediated by sex chromosomes, steroid hormones or a combination of both. Often, a particular feature under investigation is reflective of the myriad of physiological factors, genotype and experiences (psychosocial and biological) of the individual. Indeed, experiences such as childhood maltreatment or pregnancy can influence epigenetic clock ageing^{4,5}. Numerous differences can be driven by psychosocial and biological experiences conferred, in part, through epigenetic changes that do not affect XX and XY genotypes similarly^{6,7}. Indeed, X inactivation, the process by which an X chromosome in each cell becomes inactivated, results in a mosaic of populations of different cell types that can have functional differences and result in phenotype variability⁸. Although X inactivation has mainly been studied in the context of development, research shows that this process might be more dynamic throughout the lifespan than previously thought⁹. For example, Xist, a long

non-coding RNA that mediates X inactivation within each cell, is present in all XX but not in XY cells and is upregulated in human female individuals with depression¹⁰. Furthermore, research suggests that the pattern of X inactivation of genes might vary with experience across the oestrous or menstrual cycle and there are hormone-induced epigenetic modifications^{9,11}. The diversity in sex chromosome compositions (that is, XX, XXX, XY, XXY and XO) and variations in the patterns of X inactivation as well as escaped X inactivation (which occurs when some silenced X-linked genes continue to be expressed) across cells suggest that sex is better understood as a spectrum rather than as a binary category. Furthermore, sex and gender are not always aligned, as individuals who are biologically female or male (sex) do not always identify as women or men (gender), respectively^{1,12}. Moreover, individuals can have mismatching chromosomal sex, internal gonads and external genitalia at birth due to hormone receptor or enzyme deficiencies (Box 1).

The complexity of sex and gender underscores that, although challenging, studying their heterogeneity will aid in unravelling the effects of sex and steroid hormones on health and disease. However, current research practices often fall short in adequately addressing this complexity, and many studies do not accurately examine sex or gender as factors and even conflate the terms 'sex' and 'gender'³. There are ongoing discussions about best practices for investigating and reporting the multifaceted and diverse nature of sex, gender and gender-related factors^{3,13,14} (see Box 2 for examples). We encourage readers to explore other reviews on how sex chromosomes, gender and gendered variables can shape sex influences^{3,15}.

In this Review, we explore how the steroid hormones oestrogens and androgens influence brain health across sex recorded at birth. First, we discuss current challenges and opportunities in sex-informed research. Second, we synthesize current knowledge on steroid hormone effects on the hippocampus across the lifespan, with relevance to depression and Alzheimer disease. Lastly, we conclude with a discussion about the consequences of sex-biased or sex-agnostic approaches in biomedical research and how appropriately designed and conducted sex-informed research will pave the way forward towards precision medicine. Throughout this Review, we use the terms 'female' and 'male' to refer to the reported sex of the animals or human participants and the terms 'women' and 'men' as gendered terms specific to humans, if these terms were referenced in the specific literature reviewed.

Challenges and opportunities in sex-informed research

The multifaceted nature of sex differences

Sex differences are multifaceted¹⁶ (Fig. 2). Sex dimorphism is marked by completely distinct and non-overlapping characteristics between sexes (Fig. 2a). For example, male peafowl (peacocks) possess vibrant and iridescent tail feathers, which they will showcase in preparation for mating, whereas female peafowl (peahens) exhibit more subdued colours and a less striking appearance¹⁷. Although there are some clear examples of sex dimorphism in many species, more frequently, sex differences manifest as variations in a particular trait of interest, reflecting a difference in averages¹⁸. For example, human male individuals are taller on average than human female individuals, reflecting a quantitative difference¹⁹ (Fig. 2b). Human female individuals are more likely than male individuals to present with comorbid anxiety along with depression, reflecting a population level difference²⁰ (Fig. 2c). However, influences of sex can also be 'hidden', as in the case of differences in the underlying biological or physiological mechanisms (Fig. 2d) or latent differences (Fig. 2e). A sex difference in an underlying mechanism

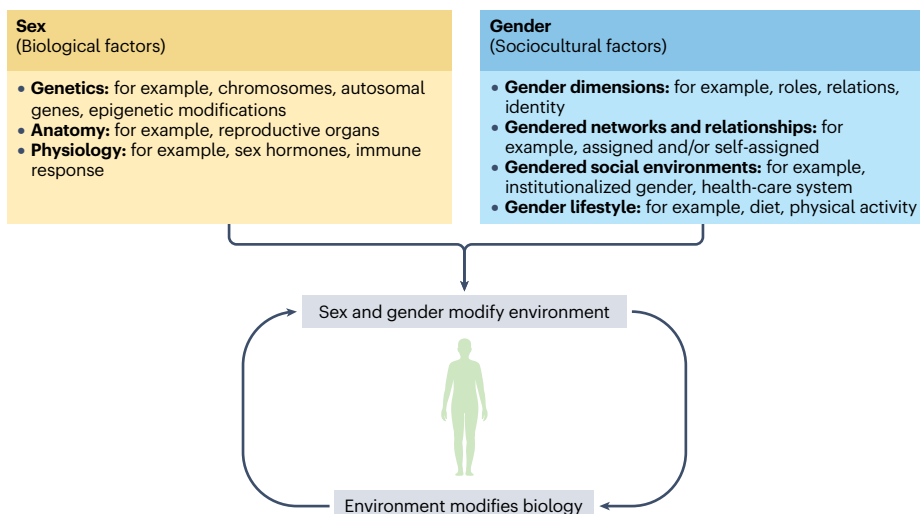


Fig. 1 | Sex and gender interactions shape health outcomes. Sex refers to biological characteristics, whereas gender is a multidimensional sociocultural construct that includes gender identities and expressions as well as socially constructed gender norms and roles. Sex and gender are often conflated in the literature³. Separating the effects of sex and gender in research can be challenging. Societal gender factors can affect biology, in part through differential stress exposure, which can impact brain health. For example, country-level gender

inequality is associated with differences in brain structure between male and female individuals¹⁹⁹. Sex is also complex, as the variability in sex chromosome composition (XX, XXX, XY, XXY and XO) and disorders of hormone sensitivity (for example, androgen insensitivity syndrome or 5 α -reductase deficiency disorder) can influence primary and secondary sex characteristics and gender identity. These examples remind us that many challenges lay ahead in unravelling the effects of sex and gender on health and disease.

can occur regardless of whether or not the trait presents itself differently in male and female individuals, and has critical implications for sex-specific treatment. For example, analysis of the transcriptional signatures associated with major depressive disorder has revealed a number of genes whose expression changes in opposing directions between male and female individuals²¹. In addition, a genetic variant, *NKAIN2*, increases the risk of schizophrenia in human female individuals but decreases the risk in male individuals²². Latent sex differences can emerge through genotype, age or disease, or following a perturbation in the environment such as stress exposure. One example of a latent sex difference involves tau, which, when hyperphosphorylated, is a neuropathological feature of Alzheimer disease. Individuals with the *APOE4* allele have higher levels of phosphorylated tau relative to non-carriers, with disproportionately higher levels in female carriers than male carriers. However, in the absence of the *APOE4* allele, no such sex difference exists²³. Finally, there are also striking differences in the metabolism of various drugs between male and female individuals, resulting in differential pharmacokinetics and considerations for treatment side effects and efficacy^{24,25}. Despite these differences in metabolism, there is still little recognition of the need for medications to be tailored by sex. Understanding sex differences, including those that are latent or present in underlying mechanisms, is imperative for identifying novel and efficacious treatment strategies that target sex-specific disease mechanisms.

Current limitations

Despite efforts to encourage the adoption of sex-based or gender-based analyses by funding agencies and publishers²⁶, progress in biomedical research remains slow^{27,28}. Indeed, it is estimated that only 5% of neuroscience studies published in 2019 incorporated sex as a discovery variable, meaning that, in only this small proportion of studies, sex was

used as a predictor variable or a between-subject variable to analyse for main and interaction effects rather than as a covariate²⁸. Using sex as a covariate removes any linear variation due to sex from the analysis and thus cannot meaningfully inform on potential effects of sex²⁹. This limited number of studies using sex as a predictor variable in analyses of outcomes has motivated considerable discussion on how best to examine, measure and analyse possible sex and gender influences in health outcomes^{30–32}.

The inadequate attention to sex differences might stem from misconceptions surrounding the study of sex differences (Box 3). Some researchers might be reluctant to analyse data using sex as a discovery variable due to concerns of insufficient statistical power. However, simulations reveal that there is no loss of statistical power in cases where there is no significant effect of sex, and power is increased where the effect of sex is in opposing directions^{33,34}. In one study, using sex as a discovery variable identified 47 asthma-associated single nucleotide polymorphisms (SNPs), compared with only 21 SNPs when sex was used as a covariate, because many SNPs contributed to the risk of asthma in opposing directions between the sexes³⁵. This result demonstrates that the failure to examine sex interactions in data analyses can result in the overall dilution of effects and, thus, in missed opportunities to discover sex-specific disease mechanisms that are vital for tailored treatments.

Another misconception is that studying sex differences is sexist. The discovery of sex differences in a trait does not imply that one sex is inferior to the other: rather, it informs on potential differences related to the underlying mechanisms of that trait. Meta-analyses indicate that sex differences in grey matter volume differ by brain region, with certain regions being larger in female individuals but others being larger in male individuals³⁶. Importantly, a small volume does not necessarily equate to inferior function, as reduced grey matter volume during pregnancy in humans is associated with improved maternal

attachment³⁷. In rats, chronic stress-induced reductions in hippocampal neuropil were associated with improved cognition in females but impaired cognition in males³⁸. It is important to exercise caution when interpreting sex differences to avoid making assumptions that imply the inferiority of one sex.

The exclusion of female individuals from biomedical research is often a result of them being deemed 'too variable' owing to their hormonal fluctuations³⁹. However, meta-analyses in rodents indicate that there are no sex differences in variability across a number of physiological traits (behavioural, electrophysiological, neurochemical, histological and non-brain measures)^{40,41}. In humans, evidence suggests a higher degree of variability in brain structure in male individuals compared with female individuals⁴². However, these findings should not be misconstrued as reasons to rule out studying hormone profiles, as what causes variability within males or within females might be due, in part, to hormones. Indeed, in some cases, examining different stages of the oestrous cycle can reveal sex differences. For example, females in diestrus, but not those in proestrus, can differ from males in anxiety-like behaviour⁴³, highlighting the importance of considering the hormonal milieu. Below, we describe how differences in steroid hormones, such as oestradiol and testosterone, differently influence hippocampal plasticity with relevance to depression and Alzheimer disease.

Steroid hormones in health and disease across the lifespan

To understand the various drivers of sex differences, it is imperative to consider the influence of steroid hormones and sex-specific physiology and experiences. In this section, we review the steroid hormones oestrogens and androgens, their receptors, their cyclic patterns across the lifespan in male and female individuals, and their influence on neurogenesis in the adult hippocampus. We then examine two disorders – depression and Alzheimer disease – that involve changes in hippocampal plasticity, focusing on the influence of steroid hormones and neurogenesis in the hippocampus in these disorders.

Steroid hormone levels

One of the fundamental biological differences between male and female individuals lies in their gonads, which release hormones across the lifespan. Both male and female individuals synthesize oestrogens and androgens, albeit in different quantities. The hypothalamic–pituitary–gonadal axis regulates the release of gonadal steroids through various feedback mechanisms. Although steroid hormones are often studied in isolation, the interplay between the hypothalamic–pituitary–gonadal axis and other axes, such as the hypothalamic–pituitary–adrenal axis, is important to consider⁴⁴. In addition to being produced and released from the gonads, sex steroid hormones are secreted by adipose tissue and can be synthesized locally from endogenous precursors (for example, testosterone to oestradiol) or *de novo* (directly from cholesterol) in the brain⁴⁵. The effect of steroid hormones in the brain is often specific to region and sex^{46,47}. There are a number of methodological considerations in the measurement of steroid hormones^{46,47} (Supplementary Box 1).

Androgen receptors and oestrogen receptors are located throughout the brain, including in the hippocampus⁴⁸. There are minimal sex differences in the distribution of these receptors, although the profiles depend on age, experience and oestrous cycle phase in rodents^{49,50}.

Oestrogens. The two primary oestrogens in humans and rodents are oestradiol and oestrone^{51,52}. In humans, oestradiol is present at

high levels pre-menopause and binds with high affinity to oestrogen receptors; post-menopause, oestrone is the most abundant of the oestrogens⁵³. Oestradiol exists as two isomers: 17 β -oestradiol (hereafter referred to as 'oestradiol'), which binds to nuclear oestrogen receptors and is extensively studied, and 17 α -oestradiol, which binds to the membrane G protein-coupled oestrogen receptor. Both oestrogens and androgens are potent neuromodulators in the hippocampus⁵⁴. The magnitude and direction of the effects of steroid hormones on the brain depend on the hormone type (endogenous or exogenous), dose, timing, age, sex and experience (for example, pregnancy), as discussed below^{55,56}. However, these factors are not often considered and are likely to contribute to some inconsistencies across studies.

Ovarian hormone cycles vary across species, with the menstrual cycle (typically seen in primates), the oestrous cycle (typically seen in rodents) and seasonal breeding cycles in induced ovulators (typically seen in voles)⁵⁷. Steroid hormones, such as oestradiol and

Box 1 | The complexity of sex

Sexual differentiation was historically thought to involve an active process in male individuals, driven mainly by the SRY gene on the Y chromosome contributing to testes development, and a default process in female individuals that occurs in the absence of these male-specific factors. Historically, female sexual differentiation has been described using language that indicates it is a passive process^{204–206}. However, it is now known to be an active process, with several genes implicated in ovarian differentiation^{207,208}. Research is lacking in terms of understanding female sex determination. Differences in sex development, also known as variations in sex characteristics, encompass a range of congenital conditions that involve altered development of chromosomal, gonadal or phenotypic sex²⁰⁹. For example, individuals with androgen insensitivity syndrome possess an XY genotype but have androgen receptor dysfunction, rendering them phenotypically female at birth²¹⁰. Individuals with complete androgen insensitivity syndrome do not possess ovaries or menstruate but have internal testes. Individuals with 5 α -reductase 2 deficiency have XY chromosomes and functioning androgen receptors, but the enzyme 5 α -reductase-, which converts testosterone to a potent androgen that is responsible for virilization during development, is impaired and thus individuals can present at birth with a spectrum of phenotypes that often appear ambiguous²¹¹. However, upon puberty, the development of secondary sex characteristics that is driven by testosterone rather than dihydrotestosterone leads to a male external genitalia phenotype. Congenital adrenal hyperplasia refers to a group of genetic disorders that affect the adrenal glands, leading to the increased production of various hormones, including androgens and cortisol²¹². Consequently, female individuals with congenital adrenal hyperplasia can present with ambiguous genitalia, precocious puberty and irregular menstrual periods²¹². These cases illustrate the nuanced nature of the presentation of sex. More research is needed to understand the active process of early female sexual differentiation, the contribution of various hormones, hormone receptor distribution and hormone receptor functioning throughout the lifespan across different genotypes as these factors will influence health outcomes in varied ways beyond binary sex.

progesterone, also fluctuate across the lifespan with pregnancy and the transition to menopause (Fig. 3). In addition, following pregnancy, ovarian hormone levels across the menstrual or oestrous cycle are lower for at least 3 years postpartum^{58,59}. Exogenous oestrogens and progestins, such as those in hormonal contraceptives, gender-affirming hormone therapy and menopausal hormone therapy (MHT), further contribute to hormonal fluctuations but are rarely considered in health research^{60,61}. Throughout this Review, we explore how fluctuations in endogenous and exogenous hormones can influence brain health and contribute to disease risk.

Androgens. In adult human male individuals, testosterone levels show diurnal variations, peaking in the morning and falling by 50% in the evening⁶². Longitudinal studies show that total and free testosterone levels steadily decline with age⁶³. Lifestyle factors, such as smoking, marital status, parental status and BMI, as well as genetic and disease-related factors can also influence testosterone levels but are rarely accounted for in the literature⁶⁴ and are likely to contribute to the lack of consensus on normative levels of testosterone in humans⁶⁵.

The ability of testosterone to influence neuroplasticity and behaviour is nuanced by a number of factors that affect testosterone signalling, including dose, androgen receptor expression and sensitivity, and the metabolic fate of testosterone. The gene encoding the androgen receptor contains a segment of CAG repeats of varied length, which affects the function and sensitivity of the androgen receptor⁶⁶. Testosterone can be converted into dihydrotestosterone, a potent androgen, or aromatized to oestradiol, thereby ultimately exerting its actions via the androgen receptor or the oestrogen receptor. Testosterone and oestradiol can act via different pathways to affect neuroplasticity in the hippocampus of male versus female individuals^{67,68}, emphasizing the existence of potential sex differences in the mechanisms underlying steroid hormone actions in the brain.

Box 2 | Best practices for the inclusion of sex and gender

- Operationalize and transparently define sex and gender within each study.
- Consider engagement of sexual and gender minority individuals to ensure an inclusive approach to research questions.
- Include sex and/or gender in demographic tables.
- Visualizations of data should differentiate between sexes and/or genders of participants (for example, using different colours) to highlight any potential contributions of sex and/or gender.
- Minimize the use of sex or gender as a covariate.
- Use sex and/or gender as meaningful or discovery variables in statistical analyses. When sample size might be an issue, consider using a sensitivity analysis rather than dropping sex or gender as a discovery variable.
- When interpreting findings, consider context specificity (species, age, genotype, hormones, environment and experience) in sex-related and/or gender-related variations in data outcomes. Assess whether the effects of gender might be driven by biological factors and/or whether the effects of sex might be driven by gendered factors.

Adult hippocampal neurogenesis: functional implications and modulation by steroid hormones

The hippocampus carries out important roles in memory and emotion and exhibits substantial plasticity throughout life⁶⁹. Different forms of plasticity are evident in subregions of the hippocampus (dentate gyrus and cornu ammonis (CA1, CA2 and CA3)), including structural changes in spine and synapse density and the production of new neurons (neurogenesis) in the dentate gyrus. The integrity of the hippocampus is compromised in several brain disorders, including major depressive disorder and Alzheimer disease⁶⁹, highlighting the importance of investigating how plasticity in the hippocampus could be targeted to reverse synapse and/or cell loss that occurs in these diseases.

Adult neurogenesis is evident in the dentate gyrus of rodents and primates. Despite a small number of studies claiming that adult neurogenesis does not occur in humans^{70,71}, a much larger body of research, using multiple methods, demonstrates robust evidence for dentate gyrus neurogenesis in adult humans⁷². Neurogenesis in the dentate gyrus is influenced by sex and steroid hormones and is reduced in Alzheimer disease and major depressive disorder in humans and in animal models; however, how these factors are linked remains to be investigated^{73–75}. Here, we focus on steroid hormone modulation of neurogenesis but we note that steroid hormones also regulate other forms of plasticity, including spine and synapse density, as reviewed elsewhere^{67,76}.

Sex differences in neurogenesis in the hippocampus. Neurogenesis comprises four different stages (proliferation, migration, differentiation and maturation), which can be examined using stage-specific endogenous and exogenous markers (Supplementary Box 2). Each specific stage can be modulated by steroid hormones. Although there are no sex differences in the overall density of new neurons in the dentate gyrus in adulthood, adult male rats show greater levels of cell proliferation, greater attrition of these new cells, and earlier maturation of new neurons compared with adult female rats, indicating sex differences in the process of neurogenesis⁷⁷. Whether sex differences in the dynamics of neurogenesis contribute to sex differences in the function of the hippocampus remains to be explored.

Adult neurogenesis is required for pattern separation, stress resilience and mediating the effects of certain antidepressants^{72,78,79}, with sex differences in all these outcomes^{80,81}. Spatial learning or pattern separation enhances neurogenesis in male but not in female rats^{82,83}, whereas contextual associative learning enhances neurogenesis in female but not in male rats in the dentate gyrus⁸⁴. Furthermore, the activity of new neurons is related to performance in sex-specific ways, as female rats show positive associations after spatial memory training, but male rats show positive associations after contextual discrimination^{85,86}. Thus, the literature suggests the intriguing possibility that there are sex-dependent networks for newly integrated neurons in the adult dentate gyrus^{86,87}. Although repeated electroconvulsive seizures, a type of electroconvulsive therapy that is used as a treatment for depression, increased hippocampal neurogenesis in male and female rats, an antidepressant-like behavioural effect was only induced in males⁸⁸. This finding demonstrates sex differences in the functional implications of modulating neurogenic-like capabilities by electroconvulsive seizures. Moreover, acute stress in adulthood or exposure to stress early in life reduces neurogenesis in adult male but not in female rats^{89,90}, with the opposite pattern exhibited when stress is experienced in middle age⁹¹. Collectively, findings suggest that new neurons are utilized differently between males and females; however, few studies to date have examined the potential differences in neurogenic outcomes across sex.

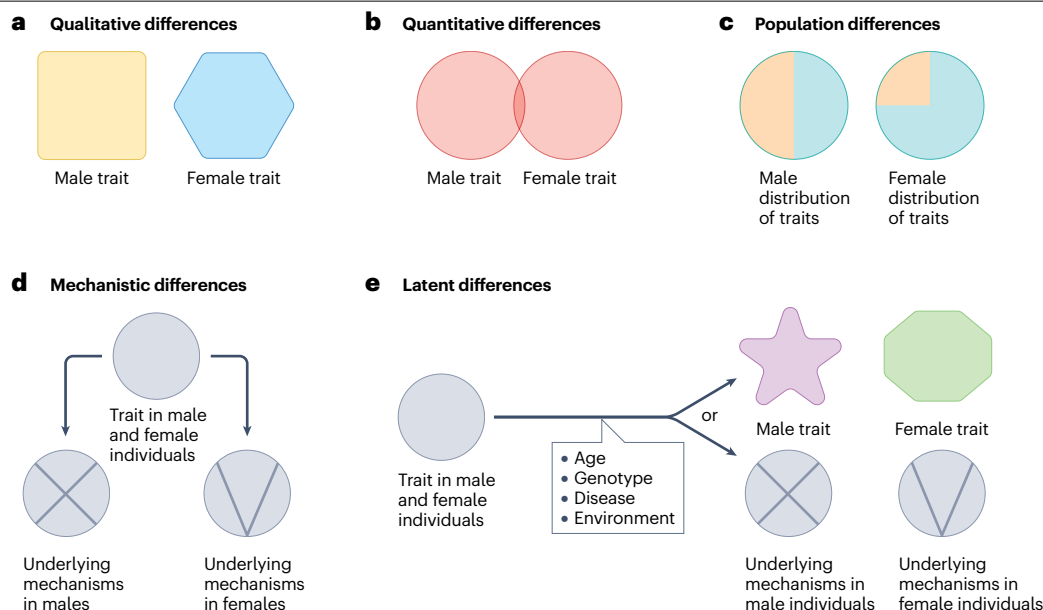


Fig. 2 | Different types of sex difference. Qualitative (part **a**) or quantitative (part **b**) differences occur when male and female individuals present different patterns of traits or a particular trait to different levels, respectively. An example of a qualitative difference is the presence of elaborate feathers in male peafowl but not in female peafowl¹⁷, whereas an example of a quantitative difference is the height of human female versus human male individuals¹⁹. Population sex differences involve a different proportion of male or female individuals presenting a particular trait (part **c**). An example of this type of sex difference occurs in major depressive disorder, in which human female individuals are more likely to present with comorbid anxiety compared with human male individuals²⁰. Hidden sex differences can include those present in underlying mechanisms (part **d**) or latent differences (part **e**). Similar traits are represented using the same shape (circles in parts **b** and **c**), and the circles in parts **d** and **e**

represent a trait that has no initial sex difference. An example of a sex difference in underlying mechanisms is provided by the *NKAIN2* gene, which is associated with a decreased risk of schizophrenia in male individuals but increased risk in female individuals²². In the case of a latent difference, a trait might not initially present with any sex differences, but certain perturbations or conditions (for example, age, disease, environment (stress) and genotype) subsequently uncover differences. For example, sex differences in the levels of phosphorylated tau, a neuropathological marker of Alzheimer disease, are not observed in the absence of the *APOE4* allele; however, individuals with one or two *APOE4* alleles have higher levels of phosphorylated tau relative to non-carriers, with female carriers having disproportionately higher levels compared with male carriers²³. Adapted with permission from ref. 16, The American Society for Pharmacology and Experimental Therapeutics.

Sex differences in the influence of steroid hormones on neurogenesis. Neurogenesis in the dentate gyrus is modulated by both androgens and oestrogens; the direction of effects depends on sex and age (Fig. 4). In rats, gonadectomy in adulthood reduces neurogenesis in both sexes, although the timing of this reduction after gonadectomy differs between males and females. In adult male rats, gonadectomy reduces neurogenesis by reducing the survival of new neurons, an effect that is only seen weeks after gonadectomy⁹². In adult female rats, ovariectomy causes an initial reduction in cell proliferation that is no longer observed 3–4 weeks after the procedure^{93,94}. Curiously, age influences the effects of gonadectomy on neurogenesis, as gonadectomy increases neurogenesis in middle-aged female rats and in juvenile male rhesus macaques, perhaps reflecting the steroid hormone milieu at the time of gonadectomy^{95,96}. These findings illustrate the complex interplay between gonadectomy and neurogenesis, which varies with age, time since gonadectomy and stage of neurogenesis.

Chronic (but not acute) exposure to testosterone or dihydrotestosterone increases neurogenesis in the dentate gyrus of adult male rats but does not alter neurogenesis levels in adult female or middle-aged male rats^{92,97,98}. Testosterone increases hippocampal neurogenesis by enhancing the survival of new neurons through an androgen receptor-dependent mechanism⁹⁹. On the other hand,

acute and chronic oestradiol exposure influences neurogenesis in both adult and middle-aged female rats but not typically in male rats¹⁰⁰. Oestradiol can either increase or decrease neurogenesis in female individuals, depending on timing, dose, age and reproductive experience¹⁰⁰.

Thus, there are numerous sex differences in neurogenesis in the dentate gyrus, including in the maturation cycle of new neurons, the activity of new neurons in response to memory, and the influence of stress or steroid hormones¹⁰⁰. How these differences might influence susceptibility to diseases that compromise the hippocampus, and how this information could be harnessed therapeutically, is not yet known.

Depression: the influence of sex and sex-specific factors

Major depressive disorder is twice as common in female individuals as it is in male individuals, and this sex difference in prevalence emerges in adolescence and remains high during the female reproductive years, pointing to a role for ovarian hormones¹⁰¹. Indeed, endogenous or exogenous hormonal transitions in female individuals are associated with an increased risk for depression in susceptible individuals^{102,103}. In this section, we consider the role of the menstrual cycle, pregnancy and the menopausal transition in female individuals, as well as the influence of testosterone on depression in male individuals.

Ovarian hormones. Fluctuations in hormone concentrations during the menstrual cycle are associated with changes in mood¹⁰⁴. Although emotion recognition does not change over the course of the menstrual cycle¹⁰⁵, reduced positive or increased negative affect are reported during the premenstrual phase¹⁰². Premenstrual dysphoric disorder, which is characterized by a cyclical emergence of mood disturbances in the luteal phase of the menstrual cycle, also presents with heterogeneity in the timing of onset between individuals. Eighty per cent of individuals with premenstrual dysphoric disorder report mood disturbances during the late luteal phase (premenstrual week), whereas 20% report mood disturbances across the entire luteal phase¹⁰⁶. This heterogeneity might reflect distinct subtypes of premenstrual dysphoric disorder; however, this subtyping is not reflected in current diagnostic manuals. Allopregnanolone, a metabolite of progesterone, is linked to mood worsening in individuals with premenstrual dysphoric disorder, and stabilizing the levels of allopregnanolone is associated with alleviation of symptoms, unlike what is seen for postpartum depression^{107,108}. Fluctuations in exogenous hormones, caused by pausing hormonal contraceptives, for example, also increase negative mood scores in humans¹⁰³. Age at initiation of hormonal contraceptive use influences factors related to the risk of major depressive disorder, with adolescent exposure increasing antidepressant use but adult exposure not affecting depressive symptoms^{109,110}. Hormonal contraceptives also influence hypothalamic–pituitary–adrenal axis dynamics, mimicking profiles of cortisol seen in individuals with major depressive disorder¹¹¹, highlighting the importance of hormonal interactions on mood.

Notably, the postpartum period is a time of elevated risk for new-onset depression in female individuals¹¹². Oestrogen and progesterone levels increase dramatically during pregnancy and decline at parturition, after which they remain low for several months¹¹³ (Fig. 3). Perinatal depression is defined as the onset of depression during pregnancy or the first month after giving birth according to the

Diagnostic and Statistical Manual of Mental Illnesses Fifth Edition¹¹⁴. The onset of depression during the postpartum period is more likely to be the first occurrence of depression, whereas depression onset during pregnancy is associated with a history of major depressive disorder, indicating that perinatal depression is a heterogeneous disorder¹¹⁵. Compelling evidence has linked the ‘withdrawal’ of steroid hormones following parturition to postpartum depression¹¹⁶. Indeed, withdrawal from a hormone-simulated pregnancy increases negative mood scores or endophenotypes in both humans and rodents^{117–119}. Allopregnanolone reverses depressive-like endophenotypes in postpartum mice, and synthetic analogues are effective therapeutics for postpartum depression in humans^{120,121}. The efficacy of allopregnanolone-based treatments for postpartum depression highlights the importance of studying sex-specific links between hormones and major depressive disorder¹²².

Lastly, the menopausal transition is another period of increased risk for major depressive disorder¹²³, and meta-analyses indicate that combining MHT with antidepressants is more effective at reducing depressive symptoms than antidepressants alone¹²⁴. Therefore, endogenous and exogenous hormonal transitions throughout the female reproductive span can stimulate negative mood.

Experimental manipulation of ovarian hormones in humans and rodents underscores the importance of these hormones in mood regulation and highlights the potential involvement of the hippocampus. In humans, withdrawal of ovarian hormones through the use of gonadotropin-releasing hormone agonists increases negative mood scores, along with reducing hippocampal volume and hippocampal functional connectivity^{117,125–127}. In rodents, withdrawal of high oestradiol and progesterone levels increases depressive-like endophenotypes, including reduced neurogenesis in the dentate gyrus, highlighting a potential mechanism linking steroid hormones and depression^{93,118,119,128}. Thus, several lines of evidence in humans and

Box 3 | Common misconceptions in the study of sex and gender

- **“Sex and gender are interchangeable terms”**

Sex refers to biological differences whereas gender is a psychosocial construct, encompassing sociocultural expectations. Differentiating between these concepts is essential for designing studies that capture the complexity of health-related outcomes.

- **“Studying sex and gender is only relevant to reproductive health”**

Sex and gender influence health beyond the reproductive system. Incorporating a sex and gender lens is crucial for understanding differences in disease prevalence, progression and treatment responses across diverse conditions.

- **“Including female individuals in research is sufficient for sex-based analyses”**

Sex-informed analyses (that is, statistically using sex as a discovery variable) require a comprehensive approach that considers male and female participants. Neglecting to examine potential sex differences could mask critical aspects of disease mechanisms and therapeutic interventions.

- **“Sex and gender are binary categories”**

Sex and gender exist on a spectrum, and individuals do not always fit into binary categories such as female or male and woman or man.

Recognizing and studying diversity in sex and gender are important for understanding health disparities and tailoring interventions to specific populations.

- **“Biological variables are the sole determinants of sex differences”**

Biological, environmental and social factors interact in complex ways to contribute to sex and gender differences (Fig. 1).

- **“Studying sex differences results in loss of statistical power”**

Rigorous study design and appropriate statistical methods can effectively address concerns about statistical power and enhance the precision and generalizability of findings.

- **“The study of sex differences is sexist”**

It is not inherently sexist to explore how biological and social factors contribute to health outcomes across sex. Research should aim to elucidate mechanisms without perpetuating stereotypes or biases.

- **“Female individuals are too variable to study”**

Variability exists in both male and female individuals. It is important to recognize and explore individual differences within each sex as these variations might provide valuable insights into health and disease outcomes.

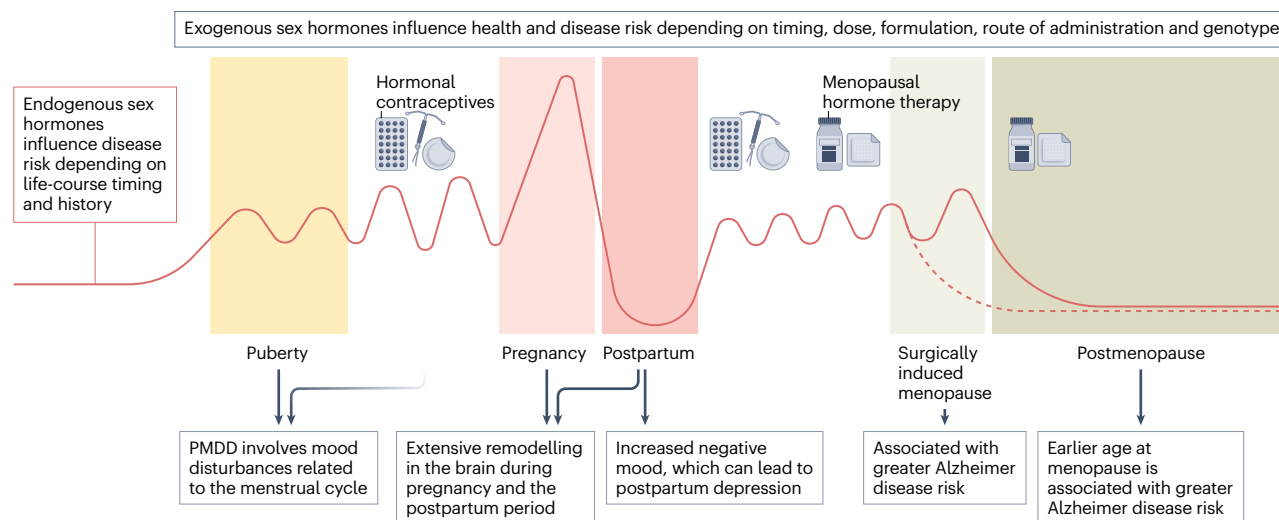


Fig. 3 | Endogenous and exogenous exposure to oestrogens and progesterone in female individuals. Starting in puberty, the ovaries release oestrogens and progesterone in a cyclical fashion in response to a hormonal cascade initiating in the hypothalamus. Hormonal contraceptives reduce the release of endogenous oestradiol and progesterone from the gonads but they contain high levels of exogenous oestrogens and progestins^{61,200,201}. Pregnancy is associated with extremely high levels of oestrogens and progesterone, including some oestrogens that are not seen outside of pregnancy, secreted by the placenta. During the postpartum period, female individuals are hypogonadal, with very

low levels of oestrogens and progesterone. Following pregnancy, long-term fluctuations of oestrogens and progesterone across the menstrual cycle are reduced. Surgical or natural menopause leads to different hormonal profiles and repercussions for ageing (dashed line versus solid line)¹⁵⁸. Each of these life stages, together with previous experiences, can influence the levels of oestrogens and progesterone to exert effects on brain health and disease risk. The effects of menopausal hormone therapy on dementia risk differ depending on formulation, timing relative to menopause and, potentially, genotype. PMDD, premenstrual dysphoric disorder.

rodents point to the withdrawal of ovarian hormones being a trigger for worsening mood.

Testosterone. Testosterone is implicated in major depressive disorder in male individuals. Lower levels of testosterone are associated with lower mood, and meta-analyses indicate that testosterone is efficacious in reducing affective symptoms of depression in both hypogonadal and eugonadal human male individuals^{129,130}. However, dose, timing, age, number of androgen receptor CAG repeats and/or interactions with other hormones such as cortisol can all be involved in moderating the effects of testosterone on the hippocampus and mood, and studies should consider these factors to uncover the influence of testosterone on mood in male individuals^{131,132}.

Sex differences in stress susceptibility to depressive endophenotypes and effects of antidepressants. Stress and female sex are major risk factors for depression; therefore, it is important to consider how steroid hormones could influence stress susceptibility versus resilience across sex¹³³. Animal models of major depressive disorder often use chronic unpredictable stress to elicit depressive-like endophenotypes, including reductions in neurogenesis in the dentate gyrus. Gonadectomy in male rats increases depressive-like endophenotypes induced by chronic unpredictable stress relative to intact rats¹³⁴. However, testosterone treatment alleviated only certain depressive-like endophenotypes in gonadectomized males^{135,136}. In female rodents, long-term ovariectomy increases depressive-like endophenotypes with or without chronic unpredictable stress^{137–139}, but oestradiol does not provide complete resilience to chronic unpredictable stress-induced depressive-like endophenotypes¹⁴⁰. Thus, although steroid hormones

are important determinants of susceptibility to depressive-like endophenotypes following chronic unpredictable stress, other factors, including hormones other than oestradiol and testosterone, are also clearly involved.

Unmedicated individuals with major depressive disorder exhibit smaller hippocampal volume and reduced neurogenesis compared with those without major depressive disorder, and these differences are particularly evident in the anterior dentate gyrus^{73,141}. Conversely, antidepressant use is associated with increased neurogenesis^{142,143}, with one study finding this effect in human female individuals but not in male individuals¹⁴³. There is also an age-related decrease in the ability of antidepressants to upregulate neurogenesis, which could be linked to gonadal hormone levels^{143,144}. In animal models, neurogenesis in the dentate gyrus is required for some but not all forms of behavioural efficacy of pharmacological antidepressants in males^{145,146}. Depleting PSA-NCAM, a cell-adhesion molecule required for the migration of new neurons in the dentate gyrus, prevented the behavioural efficacy of an antidepressant in male rats¹⁴⁷. Curiously, in female rats, chronic antidepressant treatment in combination with chronic unpredictable stress increases neurogenesis but does not alleviate depressive-like behaviours regardless of ovarian hormone status¹³⁹. These studies collectively suggest a disconnect between the ability of antidepressants to modulate depressive-like behaviour and neurogenesis in females, unlike what is seen in males. Intriguingly, evidence from a meta-analysis shows that transcripts associated with major depressive disorder were enriched for synaptic plasticity-related pathways in human male individuals but not in female individuals, suggesting that compromised neuroplasticity might be a more important feature of male major depressive disorder, whereas other processes, such as

compromised mitochondrial function, might be a more important feature of female major depressive disorder²¹. Additional research is needed to identify how hormones affect major depressive disorder biomarkers, which could lead to the identification of novel, sex-specific therapeutic targets.

Alzheimer disease: influence of sex and sex-specific factors

Alzheimer disease is a progressive neurodegenerative brain disorder that features deterioration of both cognitive ability and hippocampal integrity¹⁴⁸. The three strongest non-modifiable risk factors in humans for Alzheimer disease are advancing age, female sex and possession of the *APOE4* allele¹⁴⁹. Human female individuals experience a greater lifetime risk, more severe neuropathology and faster cognitive decline with Alzheimer disease than human male individuals¹⁵⁰. Possession of one or two *APOE4* alleles confers a 5–15 times greater risk of developing Alzheimer disease, and 60% of individuals with Alzheimer disease possess an *APOE4* allele¹⁵¹. The *APOE4* genotype confers a greater burden of Alzheimer disease in female individuals as female carriers present with a higher prevalence of Alzheimer disease, elevated levels of tau pathology, and greater cognitive decline and hippocampal atrophy compared with male carriers^{23,152}. To better understand why female individuals are disproportionately affected by Alzheimer disease, it is pertinent to examine the potential contributions of sex-specific factors such as reproductive factors and steroid hormones.

Influence of female-specific experiences. Factors such as menopausal characteristics, hormone use and previous pregnancies alter Alzheimer disease risk in human female individuals¹⁵³. Compared with premenopausal female individuals, perimenopausal and postmenopausal female individuals, on average, exhibit lower grey matter

volumes in the temporal cortex, decreased metabolic activity and increased amyloid- β deposition – all early hallmarks shared with Alzheimer disease^{154,155}. Earlier age at menopause is associated with greater tau neuropathology, particularly in female individuals with elevated amyloid- β levels, than later age at menopause¹⁵⁶. Surgically induced menopause increases the risk of cognitive impairment and reduces hippocampal functional connectivity compared with spontaneous menopause, and younger age of surgery is associated with a faster decline in memory, greater risk of Alzheimer disease and Alzheimer disease neuropathology compared with those in whom menopause was induced surgically at a later age^{157,158}. A history of many previous pregnancies is associated with an increased disease risk, although this varies by country, suggesting that intersectional factors are important¹⁵⁹. Together, this body of research illustrates that reproductive factors can alter the risk of Alzheimer disease in female individuals.

Influence of exogenous hormones: hormonal contraceptives and MHT. Exogenous oestrogens, such as past hormonal contraceptive use and MHT, also alter Alzheimer disease risk. A past history of hormonal contraceptive use is generally associated with decreased disease risk¹⁶⁰, and human female individuals with a history of MHT use have a reduced risk of cognitive decline and Alzheimer disease compared with non-users, although the risk varies with a variety of factors, including duration of use, time of initiation of MHT relative to menopause, and type of menopause (surgical or spontaneous)¹⁶¹. Although a large randomized controlled trial reported that MHT exposure increased the risk of dementia and cognitive decline compared with non-users of MHT, many trial participants had initiated MHT more than a decade past the onset of menopause¹⁶². Research indicates that MHT is more effective at reducing Alzheimer disease risk or cognitive decline when treatment

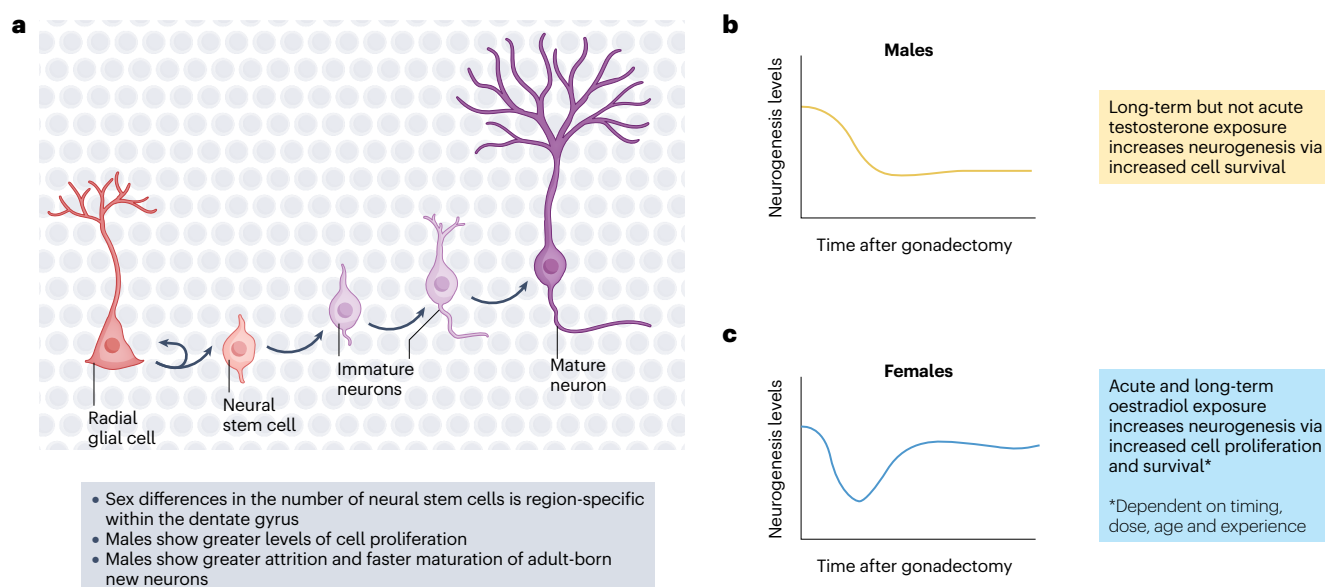


Fig. 4 | Differential effects of steroid hormones on adult hippocampal neurogenesis in males and females across the lifespan. Steroid hormones exert their effects on neurogenesis in the dentate gyrus differently in males and females, dependent on age and the stage of neurogenesis examined (part **a**). Gonadectomy reduces neurogenesis in the dentate gyrus of adult male rats (part **b**) and transiently in adult female rats (part **c**)^{92,94}. However, in middle-aged female rats and prepubertal male rhesus macaques, gonadectomy increases

neurogenesis^{95,96}. Testosterone, via androgen receptors, upregulates the survival of new neurons after long-term exposure in adult male but not in female rodents (part **b**)⁹⁷. Acute oestradiol administration rapidly upregulates cell proliferation in adult females and in reproductively experienced middle-aged female rats (part **c**)²⁰². Chronic oestradiol can upregulate or downregulate neurogenesis in adult female rodents, depending on the timing of the DNA synthesis marker injection relative to hormone treatment²⁰³.

begins within 5 years of the onset of menopause than when treatment is initiated more than 5 years after menopause¹⁶³ and that MHT benefits were more prominent when use exceeded 10 years¹⁶⁴. However, the evidence for beneficial effects of MHT remains equivocal even when accounting for time since menopause, with some studies demonstrating protective effects against cognitive decline and Alzheimer disease risk, particularly after surgical menopause, and others not¹⁶¹. The varying effects are likely to depend on other factors associated with MHT use, including duration, formulation, route of administration, menopausal symptoms and dosage, as well as the potential influence of past hormonal contraceptive use, previous parity and *APOE4* genotype^{161,165}.

Testosterone in human male individuals. Testosterone can influence cognition in older adult male individuals and affect the risk or progression of Alzheimer disease in human male individuals¹⁶⁶. Lower testosterone levels are associated with a smaller hippocampal volume, increased levels of phosphorylated tau in *APOE4* carriers, reduced cognition and higher levels of cerebral amyloid- β deposition than are higher testosterone levels^{167–169}. Animal models of Alzheimer disease demonstrate that testosterone exerts a neuroprotective effect by reducing levels of amyloid- β and increasing synaptic plasticity in the hippocampus¹⁷⁰; curiously, however, testosterone does not enhance neurogenesis in middle-aged male rats⁹⁷. Readers are directed to other reviews for a more detailed discussion of testosterone and Alzheimer disease¹⁶⁶.

Lessons from animal models. Evidence suggests that, as the severity of Alzheimer disease progresses, neurogenesis in the dentate gyrus declines in humans⁷⁵. There are numerous animal models of Alzheimer disease, each with advantages and disadvantages¹⁷¹. In several mouse models, including *APOE4* knock-in mice expressing humanized *APOE4* (*hAPOE4*) and 3xTg-AD mice, neurogenesis is reduced compared with middle-aged wild-type mice^{172,173}. Although relatively few studies have been conducted to examine the effects of steroid hormones in these models, steroid hormones have been found to influence neurogenesis. Ovariectomized APP/PS1 mice (a double-transgenic mouse model of Alzheimer disease) show an initial increase in cell proliferation followed by a progressive reduction in the number of new neurons starting from 6 months of age compared with wild-type mice, and treatment with oestradiol shortly following ovariectomy reversed these effects¹⁷⁴. This observation is consistent with other work showing that chronic administration of oestradiol early in Alzheimer disease pathology increases hippocampal neurogenesis and ameliorates cognitive performance in A β 1–42 female mice¹⁷⁵. Other reproductive factors, such as pregnancy history, influence neuroplasticity and neuropathology in animal models of Alzheimer disease. In wild-type rodent models, previous parity is associated with increased neuroplasticity and cognition but greater disease neuropathology in APP23 female mice¹⁷⁶ or with reduced neurogenesis and activity of new neurons in *hAPOE4* female rats⁷⁴. These findings could shed light on conflicting findings of previous parity in humans as the effects of parity differ by genotype and risk for Alzheimer disease. These discoveries highlight the value of investigating sex and sex-related variables to understand disease endophenotypes.

Considerations in sex-informed research

Within-sex considerations

The literature presented in this Review underscores the important contributions of sex and steroid hormones to hippocampal neurogenesis,

Box 4 | Intersectionality in health research

Intersectionality is a term that is broadly used to describe how various social and biological factors can interact to affect health that cannot be sufficiently understood by examining each identity in isolation²¹³. Biomedical research has a diversity data gap that adversely affects the health of historically under-represented groups such as women and gender-diverse individuals²¹⁴. Social intersectional variables, which represent the complex interplay between a person's identities in relation to social systems of power and oppression, are often ignored²¹⁵. The majority of researchers do not collect data on the diversity of their sample beyond age and binary sex. Sex can interact with various intersectional factors, including gender and sexuality, to affect health outcomes, and consideration of these factors, from data collection to interpretation of results, is vital^{3,216}. The importance of intersectionality in brain health research can be seen in the case of transgender health. Relative to cisgender individuals, transgender individuals are at a higher risk for major depressive disorder²¹⁷ and Alzheimer disease²¹⁸. This heightened risk is often explained by the minority stress model, which stipulates that stigma, prejudice and discrimination of marginalized communities create a stressful social environment that increases risk for mental health problems^{219,220}. For many transgender individuals, gender-affirming hormone therapy reduces symptoms of anxiety and depression, improving quality of life and self-esteem²²¹. Dosage, formulation and age at initiation of gender-affirming hormone therapy relative to puberty are important aspects to consider when studying the long-term effects of gender-affirming hormone therapy on brain health²²². These types of study will inform on how exogenous hormones influence brain health and can be extrapolated to improve health across the entire gender spectrum.

major depressive disorder and Alzheimer disease. An exclusive emphasis on sex differences falls short of providing a comprehensive understanding of how sex-specific factors and experiences can drive disease onset and progression, which will be critical for improving health outcomes for all. Furthermore, most of the literature is fixated on the binary effects of sex or gender. Unfortunately, little research has been dedicated to understanding how female-unique experiences, such as pregnancy and menopause, can influence the brain health of women^{28,61}. Moving forwards, applying a revised intersectional framework is key to understanding the multitude of social and biological factors that can contribute to the effects of exogenous hormones (such as hormonal contraceptives, gender-affirming hormone therapy or MHT) on brain health (Box 4), and is crucial to achieving equitable and effective health care.

Female-specific experiences. As outlined in this Review, female-specific hormonal experiences influence the risk for diseases such as Alzheimer disease and major depressive disorder. Variables relating to each of these hormonal events (for example, menopause type, length of menopause transition, number of previous pregnancies, menstrual cycle duration and length) and exogenous hormone treatments (for example, formulation, duration of exposure, timing, route and cyclicity) are important to consider in order to resolve conflicting

data and gain novel insights into contributions to health and disease. With the welcome influx of large databases to help to uncover health associations across many individuals, it is vital to include these variables to gain a comprehensive understanding of how endogenous and exogenous hormone exposures influence brain health. Indeed, medroxyprogesterone acetate, a synthetic progestin often used in hormone therapies, is associated with negative outcomes for dementia, ischaemia and cerebral oedema in human female individuals¹⁷⁷, yet databases often lack information on the formulation of such therapies, making appropriate analyses impossible. We urge the research community to include such details, as doing so might lead to critical, life-saving discoveries in precision medicine.

Another theme that emerges is that female-specific mood disorders, such as premenstrual dysphoric disorder and perinatal depression, are heterogeneous in terms of timing of symptom onset, coinciding with vastly different hormonal states. However, diagnostic manuals often do not distinguish between these subtypes of disorder, which is a missed opportunity to uncover how the differing aetiologies might have implications for optimal treatment strategies. Indeed, allopregnanolone has demonstrated efficacy for treating postpartum depression but not premenstrual dysphoric disorder, which could be attributed to differences in dosage and/or the hormonal milieu¹⁰⁸. In addition, major depressive disorder and how different subtypes of major depressive disorder interact with each other are not well researched but have implications for risk prediction and early intervention. For example, female individuals who show mood sensitivity to hormonal contraceptives are at increased risk for postpartum depression¹⁷⁸, and a diagnosis of major depressive disorder in female individuals is associated with an increased risk of a depressive episode during pregnancy as well as of premenstrual exacerbation of depressive symptoms¹⁷⁹. Thus, taken together, these studies suggest that certain female individuals might exhibit greater susceptibility than others to mood alterations with hormonal changes, which can inform on risk and treatment strategies. These findings also underscore the importance of studying female-specific differences for understanding the different subtypes of major depressive disorder.

Male-specific experiences. Although female-specific experiences, such as pregnancy and menopause, have formed a large focus of this Review, male-specific experiences are also important to consider to understand how hormones affect brain health. Testosterone levels are sensitive to different experiences in males. Fatherhood is correlated with reduced circulating levels of testosterone as well as changes in brain structure and plasticity, including the hippocampus, in humans and biparental rodents^{180–182}. In rodents and primates, fatherhood is associated with increased spine density and neurogenesis in the hippocampus, effects that might be mediated by changes in testosterone levels^{180,183}. Fatherhood is also associated with an increased risk of depression in humans but the heightened risk occurs later in the postpartum period, in contrast to an earlier period of risk in postpartum female individuals¹⁸⁴. Having four or more children or having no children is associated with a higher risk of Alzheimer disease in male individuals compared with having one to three children; a similar association between the number of children and the risk of Alzheimer disease was seen in female individuals¹⁸⁵. Although this effect has been attributed to lifestyle and psychosocial factors, the alteration of hormone levels associated with fatherhood suggests that hormonal contributions to the increased risk of Alzheimer disease in male individuals are also plausible.

What do we risk missing when sex is overlooked?

It is crucial that we assess whether conventional ‘gold standard’ measurements and thresholds have been established using data derived only from one sex. Neglecting to do so introduces the risk of biasing our data and obscuring the possible discovery of precision therapeutics. Indeed, there is growing recognition of the need to consider sex-specific thresholds for physiological measures in heart disease in humans¹⁸⁶ as well as for analyses of fear and cognitive behaviours in animal models^{187,188}. The diagnostic criterion of amnesic mild cognitive impairment can serve as an example of the importance of considering sex-specific thresholds. Approximately 50% of individuals with amnesic mild cognitive impairment transition to Alzheimer disease; therefore, accurate diagnosis could enhance the prediction of Alzheimer disease transition¹⁸⁹. However, human female individuals score higher than male individuals on the memory tests that are used in the diagnosis of Alzheimer disease¹⁹⁰. When sex differences in memory performance were accounted for, 10% fewer male and 10% more female individuals were diagnosed with amnesic mild cognitive impairment than when non-sex-specific diagnostic criteria were used¹⁹¹. The consequences of misdiagnosing 20% of cases are considerable as accurate diagnoses are critical for early intervention and research progression. Furthermore, misdiagnoses will hinder the discovery of biomarkers associated with the risk of Alzheimer disease transition. This example starkly illustrates that our knowledge of various health and disease outcomes will be constrained and hampered without examining the potential influences of sex.

The diverse examples of sex influences on health outcomes outlined in this Review underscore the multifaceted nature of sex and sex-related variables. It is necessary to emphasize that a lack of a sex difference from the initial examination of an outcome does not negate the possibility of latent sex influences or sex differences in underlying mechanisms that require more thorough investigation and nuanced frameworks to uncover and understand. As previously mentioned, in 2019, only 5% of published papers across six journals in the area of neuroscience statistically investigated whether sex made a difference in the findings, indicating that few studies are exploring whether sex influences outcomes²⁸. Another analysis found that only 20% of all papers in human psychiatric neuroimaging published prior to 2020 examined sex as a statistically meaningful variable¹⁹². However, of these 50 published papers, 72% found that sex affected the outcome, demonstrating the prominent effect of sex on the neurobiology of psychiatric disorders¹⁹².

Adding another layer of complexity, gender-related variables can contribute to the manifestation, expression and diagnosis of health conditions in various ways, both independently and in convergence with sex-related variables³. For example, human female individuals derive greater benefits from exercise, with respect to all-cause and cardiovascular mortality risk reduction, than male individuals at lower levels of self-reported exercise¹⁹³. However, as indicated by the authors of the study, gender-related variables make it difficult to make accurate conclusions, as non-formal exercise, such as household chores, was excluded in self-reported exercise¹⁹³. Human female individuals are more likely to be older and report more severe pain and disability once selected for joint replacement surgery than male individuals¹⁹⁴, which might be due to female individuals reporting more adverse health-care provider interactions and dismissal of symptoms than male individuals¹⁹⁵. These examples illustrate that both biology and gendered variables influence outcomes.

Lastly, it has been argued that findings of sex-related differences in brain volume perpetuate gender stereotypes and that these sex

differences in volume are actually eliminated when considering sex differences in brain or body size¹⁹⁶. However, structural volume differences are still seen across many brain regions in humans, some favouring female individuals and some favouring male individuals³⁶, even after consideration of brain size differences¹⁹⁷. Furthermore, a smaller volume in structure does not denote inferior function of that region – indeed, lower volumes of cortical areas of human brains after pregnancy are related to attachment³⁷. Sex differences in the brain are not limited to structural volume but are also seen in molecular features, gene expression patterns, functional connectivity, receptor concentrations and neurochemistry¹⁹⁸. It is important to interpret findings of sex differences in the brain with care such that their relevance is not overstated and to not assume that these are related to the inferiority of one sex compared to another¹⁹⁹. Accordingly, best practices (‘roadmaps’) for the appropriate consideration of sex-related variables in structural volume as well as the presentation and interpretation of findings are available³¹.

Conclusions

Despite concerted efforts globally, research has yet to realize the promise of precision medicine. Most scientific discoveries have stemmed from the study of male members of model species only, establishing male physiology as the norm, with serious health consequences for other sexes and genders. Given the sheer number of ways in which sex can influence health outcomes, it is critical to not mistake an initial ‘lack of sex difference’ in a given trait as evidence of no further sex differences in underlying mechanisms or other conditions. Researchers are urged to consistently incorporate sex into study designs and to analyse data appropriately in order to meaningfully examine sex as a variable²⁸. Once a sex influence is found, researchers are encouraged to consider the extent to which it might be driven by hormones, age, genotype and/or intersectional factors. Throughout this Review, we have outlined examples of how hormonal transitions across the lifespan can influence disease risk and hippocampal plasticity, highlighting the need to study within-sex factors to aid in discovery.

Although uncovering potential sex and gender differences is important, it is crucial not to focus exclusively on comparisons across sex and/or gender. By doing so, we risk defaulting to the standard of one sex (typically male), leading to the characterization of other presentations as ‘atypical’, which has repercussions for diagnoses and research outcomes. As such, it is vital to consider within-sex comparisons to shed light on how experiences unique to each sex can influence health outcomes and disease risk. Overall, the prevailing reliance on male-centric studies, oversimplification of sex and gender, and lack of attention to within-sex considerations critically overlook the variations in health outcomes and disease mechanisms across different sexes and genders. To truly achieve personalized and effective health care, research must thoughtfully incorporate sex-specific and gender-specific factors and be rigorously designed, conducted and analysed. This commitment will ensure that medical advancements are equitable and tailored to the diverse experiences of every individual.

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